

PRE-CLASS READING 1 OF 4

The bottleneck is the operating model — the AI factory, and the claim that scale never stops paying

On Iansiti, M. and Lakhani, K. R. "Competing in the Age of AI." *Harvard Business Review*, January–February 2020 — the article version of the argument in Chapters 1–2 of their book *Competing in the Age of AI: Strategy and Leadership When Algorithms and Networks Run the World* (Harvard Business Review Press, 2020). **This brief is the assigned reading; the article and the book are optional.** This is the conceptual spine of Week 5, and its flagship example is the company in our Capsule-3 case.

ASSIGNED READING

Read this brief in full. It is the assigned reading and it is self-contained — nothing in it or in class assumes you have seen the original. The HBR article sits behind a paywall, and the licence under which the University subscribes to it does not permit us to circulate a copy or to post a link for coursework, so we have not made it a requirement. Read once for the argument, then a second time asking only one question: **which claims are about technology, and which are about how firms are organised?** The second set is the durable half.

If you want the original — and you should, if the argument grabs you: the article is nine pages and NTU students can read it through the Library's **Business Source Complete** database (search the title; some HBR articles there are read-on-screen only). The book's Chapters 1 ("The Age of AI") and 2 ("Rethinking the Firm") carry the same argument at length and are worth buying or borrowing. Neither is examinable. **This brief goes further than the article in one respect:** the article is from 2020 and does not know about per-token pricing, so the section "Where the argument has aged" below has no counterpart in the original.

» **this brief ~26 min • optional original article ~15 min**

Note on currency. This was published in **January 2020** — before ChatGPT, before foundation models were a purchasable input. Every AI system in it is a predictive model a firm trained on its own data. Two things that changes. **(1)** The model layer is now largely **rented**, not built, which collapses the entry cost the book treats as a moat — but the data pipeline, which is the hard part, is exactly as hard. **(2)** Their hero example, **Ant Financial**, had its record-breaking IPO cancelled by Chinese regulators ten months after the book shipped — and it is our Capsule-3 case. Read the argument for its *structure*; treat its 2019 examples as evidence with a date on them.

KEY TAKEAWAY

A firm has a **business model** — what value it promises and how it gets paid — and an **operating model** — how the work actually gets done. Iansiti and Lakhani's claim is that most strategy attention goes to the first while the second is what sets the limits, and that AI changes the second so fundamentally that the old limits stop applying. In a traditional operating model, value is delivered by people executing processes: to serve more customers you hire, and every hire adds coordination cost, so **returns to scale eventually flatten**. To serve a different customer need you need different specialists, so **scope is expensive**. And the firm improves only as fast as its people learn. In a **digital operating model**, the work is done by software running on data. Scale becomes a deployment question, scope becomes an interface question, and the system learns from every transaction it processes. Hence their headline claim: *"the value of scale eventually begins to level off in traditional models; in AI-based ones, it never stops climbing."* The engine that does this they call the **AI factory** — a repeatable production line that turns data into decisions. Their evidence: **Ant Financial reached over a billion consumers within five years of launch, with fewer than one-tenth the employees of the largest US banks, and was valued at \$150 billion in its 2018 round — roughly half of JPMorgan Chase — while operating across payments, consumer lending, wealth management, insurance and credit scoring at once.**

WHY IT MATTERS FOR THIS COURSE

Weeks 1 to 4 were about what the technology can do. Week 5 is about where it is worth doing, and this reading supplies the frame we will use all session. In **Capsule 1** you will select and rank use cases — and the framework you rank them with is this one: does this use case buy **scale**, **scope** or **learning**, and does our operating model let us collect the benefit? In **Capsule 2** you will build a cost-to-serve calculator, which is the arithmetic that decides whether their "never stops climbing" claim is true for *your* use case in 2026 (Reading 3 argues it usually is not, at least not automatically). In **Capsule 3** we discuss Ant Group — the book's exhibit A — and the discussion turns on a bottleneck this framework does not have a slot for. Bring the framework, and bring your scepticism about it.

Feeds Class 5: C1 (use-case selection • the AI factory • network effects and data flywheels • build-vs-buy at portfolio level) • C2 (the marginal-cost claim, tested with real prices) • C3 (the Ant Group case, which is this article's own example).

MAIN POINTS

The distinction that carries everything: business model vs operating model

- **Business model = the promise and the payment.** What you sell, to whom, at what price, and why they choose you. Almost all strategy teaching lives here.
- **Operating model = how the work gets done.** People, processes, systems, and the way capacity is added when demand grows. It is the unglamorous half, and it is the half that sets your ceiling.
- **The argument in one line.** Two firms can hold the *same* business model and behave completely differently, because one adds capacity by hiring and the other adds capacity by deploying. Ant and a retail bank both promise “lend money to people who need it”; they are not in the same business at all once you look at how the promise is kept.
- **Why this is the right lens for an AI course.** “Should we use AI here?” is almost never a model question. It is a question about whether your operating model can *absorb* a model’s output — whether there is a decision it can make, a system it can write to, and a measurement that tells you whether it helped.

The AI factory — four components, and why each is needed

They call it a factory deliberately: a factory takes a raw input and produces a manufactured output on a repeatable line, at a known cost per unit. Here the raw input is data and the manufactured output is a **decision**. Four components, each of which fails the whole line if it is missing.

- **1 · The data pipeline.** Gathers, cleans, integrates and makes data usable — **systematically, sustainably and at scale**, not as a one-off extract. *Why it is needed:* a model is a function from features to a prediction, and it can only be as timely, complete and correct as the feature it is handed. *What it achieves:* it turns data from an asset someone owns into an input the line consumes every second. This is the component that takes years, and the one you cannot buy.
- **2 · Algorithm development.** The models that generate predictions about future states or actions. *Why it is needed:* the pipeline moves data; something has to convert it into a judgement. *What has changed since 2020:* for a large class of problems this component is now **rented from a vendor by API call**. That is the single biggest change to the book’s economics, and Reading 3 puts a number on it.
- **3 · The experimentation platform.** The mechanism for testing hypotheses — that changes to the model or the product actually cause the outcome you wanted. *Why it is needed:* without a controlled comparison you cannot tell improvement from drift, seasonality or luck. *What it achieves:* it converts opinion into evidence at a rate the organisation can afford. This is the same argument as Week 3’s eval set, one level up: **you cannot improve what you are not measuring, and you cannot attribute what you are not comparing.**
- **4 · The software infrastructure.** The plumbing that makes the other three consistent, modular and available to whoever needs them — internally and, increasingly, externally through APIs. *Why it is needed:* it is what stops the factory being one clever team’s private tool. *What it achieves:* **scope**. Once payments data sits behind an interface, the lending team, the insurance team and the wealth team can all draw on it without asking anyone’s permission.

Read the four in that order and notice the shape: the component you cannot rent is the first one. In 2026 anybody can have component 2 by lunchtime. Almost nobody can have component 1 by Christmas.

Worked example — the same loan decision, two operating models

A mid-sized bank underwrites unsecured personal loans. A credit officer reads an application, checks documents, applies policy and decides. Assume a good officer clears **six applications a day**; assume fully-loaded cost of about **S\$400 a day**, so roughly **S\$67 per decision**. To double volume you hire a second officer, then a supervisor for every eight officers, then a quality-assurance function to keep the two branches deciding the same way. Cost per decision does not fall as you grow; past a point it rises, because coordination is the thing you are buying. That upward drift is exactly what “returns to scale level off” means in practice.

Now the digital operating model. A scoring model reads a feature vector — transaction history, repayment behaviour, merchant categories, device signals — and returns a decision. The **marginal cost of the ten-thousandth decision is essentially the same as the marginal cost of the tenth**, and it is small. The case we discuss in Capsule 3 states the endpoint plainly: Ant’s MYBank ran each SME application through **3,000 risk-control strategies**, approved in **under one minute, around the clock, with zero human intervention**. And across Ant’s lending as a whole in 2019, the case reports a delinquency rate of about **2% against roughly 6% for banks**. Note carefully which number is the impressive one. Faster and cheaper is the operating-model story. **Lower defaults at the same time is the learning story**, and it comes from the pipeline, not the algorithm: every repayment is a labelled training example arriving for free.

The honest part of the example is what it cost to get there. The scoring model is a few weeks of work. The pipeline that puts a clean, timely, permissioned, consented feature vector in front of it — sourced from a payments network the firm happens to own — is a decade. **When someone tells you AI made a process 100× cheaper, ask them how long the data plumbing took, and who already owned the data.**

Scale, scope, learning — the three things a use case can be buying

- **Scale.** More of the same decision, at a marginal cost that stops tracking volume. *Test:* take your **current** operating model to ten times today's volume and ask two questions — what happens to the cost curve, and what breaks first? There are three honest answers. **Superlinear:** cost rises faster than volume, because you cannot hire ten times the underwriters at the same quality of judgement — supervision, error rates, queueing and training lead time all pile on top of headcount, and the model usually stops *working* before it stops being affordable. This is where the prize is largest, and it is the case most often under-claimed. **Linear:** one person per N units, ten times the volume for ten times the cost. The common and perfectly respectable case; automating the unit decision creates real value. **Already flat:** deterministic software handles it today, so scale is not what you are buying. Failing the test is not disqualifying — it relocates the argument to scope, to learning, or to a decision software could not make.

Two things the scale test does not tell you, and both of them sink real projects. **Whether ten times the volume exists.** A process that runs forty times a month has no scale value however steep its cost curve, because the fixed monthly floor swamps it — Reading 3 puts numbers on exactly that. And **whether the binding constraint at ten times is money at all.** If what breaks first is judgement rather than cost, AI relocates the bottleneck to review rather than removing it: you buy throughput up to the point where a human still has to sign, and not one unit further.

- **Scope.** The same capability redeployed into an adjacent problem. *Test:* does the asset you build for this use case have a second customer inside the firm? Ant's payments data made lending possible; the lending data made insurance possible. Each new business arrived cheaper than the last, which is the whole point.
- **Learning.** The system gets better because it ran. *Test:* does using the system produce **labelled outcomes** that flow back in? A chatbot that nobody rates learns nothing. A loan book learns automatically, because default is a label the world supplies for free. **The presence or absence of a free label is the single best predictor of whether a use case compounds.**
- **Network effects sit on top of all three, and are not the same thing.** A *direct* network effect: the product is better because other users are on it (Alipay is useful because merchants accept it). An *indirect* or **data** network effect: more usage makes the *predictions* better for everyone. The second is the one this article is really about, and it is the one most often claimed without evidence.

Collisions — why the incumbent loses without being outcompeted at its own game

- **The claim.** When a firm with a digital operating model enters a traditional industry, the two models “collide”, and the collision is asymmetric: the digital entrant is not better at banking, it is differently constrained. It can serve segments the incumbent's cost structure made unservable.
- **In this case, literally the unbanked.** Ant's growth came from customers a branch network could not profitably reach — the case reports about **72% of Chinese adults financially illiterate** in the mid-2010s and traditional credit reporting covering only about **25% of the population**. The incumbent had not decided to ignore them; its cost per decision made them unreachable.
- **The uncomfortable corollary for incumbents.** You cannot answer a collision by buying a model. The entrant's advantage is structural, so the response has to be structural — which is why “we hired a Chief AI Officer” is usually not an answer, and why Reading 2 finds that the firms reporting EBIT impact are disproportionately those that **redesigned workflows** rather than those that bought tools — a correlation Reading 2 then takes carefully apart.

WHERE THE ARGUMENT HAS AGED — READ THIS BEFORE YOU USE THE FRAMEWORK

The house habit in this course is to give a source a hard time before adopting it. Four places where the 2020 argument does not survive contact with 2026, each of which we will use in class.

1. **“Marginal cost approaches zero” is now false in an important way — and the cause is worth being precise about.** It is not a pricing accident. It is **where the expensive computation sits relative to the sale**. In classical software the costly thinking is done **once, before any customer exists**, and frozen into code; serving the next customer replays that frozen work in microseconds of CPU time, so cost per request is a rounding error and gross margins run **80–90%**. A large model replays nothing. It **re-derives the answer from scratch on every single request** — a forward pass through hundreds of billions of weights held in scarce GPU memory — and that pass is metered, in tokens, every time. The intelligence is never frozen into the binary, so it never stops being paid for. Three things sharpen it: the cost is not one number but a **spread of roughly 20–45× across model tiers** for the same task; **failure has a price too**, because a wrong answer is picked up by a person, and Reading 3 shows that fallback term running to **80–99% of the cost of a resolved task** — a line classical software has no analogue for at all; and in agentic use one request becomes many calls whose context grows each turn. Hence AI-native gross margins around **50–60%** against classical software's 80–90% (Bessemer, February 2026). Be careful what the criticism claims: inference prices have fallen steeply and keep falling, and caching, distillation and model routing all pull margin back up. The durable point is narrower and more useful — **marginal cost is no longer approximately zero, so it now has to be actively managed, and classical software never had to manage it**. The scale curve did not disappear; it acquired a floor. Reading 3 is entirely about that floor.

2. **The data moat weakened where the model is rented — which leaves the pipeline as most of the moat.** The book's implicit moat is "we have the algorithms and the data to train them." When any competitor can call a frontier model for the price of a coffee, the algorithmic half of that moat is gone, and what remains is proprietary data, distribution and workflow integration. Narrower than the book implies — and precisely why the pipeline component matters **more** now, not less. **You have already met both ways of acquiring one.** Ant's data layer arrived as a **by-product**: it was running a payments network, and creditworthiness fell out of the exhaust. **DBS paid for its layer deliberately** — our Class-2 case is an IBM Watson experiment killed in 2014 at about 80% accuracy, followed by years of unglamorous consolidation of **twelve separate data warehouses into one platform, ADA, by 2018**, with the governance framework built alongside it, before any of the generative work on top became possible. Different countries, different starting points, the same decade — Singapore and China, one bank and one payments platform. **Neither firm rented it, and no API price list shows it.**
3. **Regulation is treated as an externality, and it is not.** The operating-model framework treats every bottleneck as *internal* — hiring, coordination, specialisation — and predicts that removing them lets the firm keep climbing. It has no slot for a bottleneck imposed from outside precisely *because* the firm climbed. Our case is the book's own exhibit A being stopped, two days from listing, by exactly such a bottleneck. That is not a footnote; it is a missing term in the framework. **And do not file this under "China"**: social-media recommendation met content and platform-transparency duties without any capital rule, and digital advertising was reshaped not by a regulator at all but by **Apple**, when it changed its tracking rules in 2021. **The missing term is any externally imposed constraint that arrives because you succeeded — regulator, platform, or public.** Worked through in the Q3 notes below.
4. **The evidence is selected exemplars, not a sample.** Ant, Amazon, Google, Netflix, Ping An — firms chosen because they succeeded, then examined for what they had in common. That is a real finding about *mechanism* and a weak finding about *frequency*: it cannot tell you what share of firms that build an AI factory get these results. Reading 2 supplies the frequency, and the frequency is sobering.

You have already run the experiment behind the first of those four. In **Class 1** you trained a classifier on ten thousand labelled reviews, then put the identical question to a foundation model with **zero** training examples, and the comparison table you printed carried the row that is this whole argument: cost per 1,000 predictions, **~\$0.00** for the classifier you had trained against **"real money, every time"** for the API. Look at where the cost sat in each. For the classifier, **training** was the expensive step — paid once, in labelling effort and a few seconds of fitting — and **inference** was then free forever, because the weights were yours, they sat in your notebook's memory, and a prediction took a fraction of a millisecond of your own CPU. For the foundation model there was **no training step for you at all**, and every inference was a metered round trip to weights you do not own, on a GPU you do not own. The notebook closed on **"one needs your data, the other needs your money"** — and Iansiti and Lakhani were writing in a world where only the first kind existed. That is the entire delta. Capsule 2 puts a price on it.

READING FOCUS QUESTIONS — bring your answers to class

There are no wrong answers here — these surface how you think, not what you recall. A clue sits under each question to give you a direction to push on; the fuller reasoning is in the section after.

- Iansiti and Lakhani say the value of scale "never stops climbing." Every inference in 2026 costs money. Does the claim survive — and if it survives in a weaker form, what is the weaker form?

Clue. Do not argue this in words — split the cost into **fixed** and **variable** and do the arithmetic. Take one million users, one call each, roughly 1,000 input and 300 output tokens per call, and price it twice: at a cheap tier (about **\$0.10 per million input / \$0.40 per million output**) and at a frontier tier (about **\$5 / \$25**). Two numbers, one page apart. Then ask which of the AI factory's four components has a marginal cost that really is near zero, and which one you just priced.
- Of the four AI-factory components, which would you build first in an organisation you know — and which is the actual bottleneck there? Say what makes you sure they are not the same component.

Clue. Rank them by what you **cannot rent**. For each component write one line: can we buy this tomorrow, in a quarter, or not at all? Then notice that the one you build first is usually chosen by what is easy to start, while the bottleneck is chosen by what is hard to finish — and that gap is where most AI budgets go to die.
- The article's flagship example had its IPO cancelled by regulators ten months after publication. Does that falsify the operating-model argument, confirm it, or fall outside it?

Clue. Resist picking a side. Separate two different claims: "a digital operating model removes the internal limits on scale" and "therefore the firm keeps growing." Ask which of those the cancelled IPO is evidence about. Then ask the more interesting question — whether a bottleneck that appears because you removed all the others is genuinely outside the model, or is the model's own prediction with the sign flipped.
- Take one AI use case in an industry you know. Is it buying **scale**, **scope**, or **learning**? Be exact — most people claim all three and can defend none.

Clue. Use the three tests from the Main Points, in order, and let them be self-selecting. Scale: at 10× volume does the cost curve stay flat, go linear, or bend upward — and does 10× volume actually exist? Scope: does the asset have a second internal customer? Learning: **does using it produce a free, correct label that flows back in?** The third test is the one that eliminates most candidates, and “we will have humans rate the outputs” is a cost, not a free label.

- If the model layer is rented and your competitor rents the same one, where does durable advantage actually live? Name three places, and rank them by how long a competitor would need to copy them.

Clue. Build the list the other way round: inventory everything a well-funded competitor could have **by next Tuesday** — the model, the framework, the cloud, the prompt, probably your public documentation — and whatever survives that cull is the candidate list. Then time-cost each survivor in months. Notice that the winners are usually boring: proprietary data with consent to use it, distribution, a workflow people are already inside, and a regulatory permission that took three years to obtain.

HOW TO THINK ABOUT THESE — read after you have had a go

These are not model answers to be memorised — they are the line of reasoning we will pick up in class. Class time is short, so they are written to stand on their own.

Q1 — The claim survives, but only for the fixed half

Do the arithmetic first. One call of 1,000 input plus 300 output tokens costs $\$0.0001 + \$0.00012 = \$0.00022$ at the cheap tier and $\$0.005 + \$0.0075 = \$0.0125$ at the frontier tier. A million users, one call each: **about \$220** versus **about \$12,500** — a 57× spread for the same feature. Neither number is zero, and both grow **linearly** with users. So the strong form of the claim — marginal cost approaches zero, therefore scale pays forever — is dead in 2026, and it died because the algorithm component moved from a fixed cost you amortise to a variable cost you incur. The weaker form is still true and still important: **the fixed components of the AI factory — pipeline, experimentation platform, infrastructure — are built once and amortised over every decision**, and it is there that scale still compounds without limit. This gives you a genuinely useful decomposition to carry into Capsule 2. Total cost = a large fixed investment that gets cheaper per unit forever, plus a variable per-call cost that never does. Your job as a designer is to push work from the second bucket into the first — caching, smaller models, retrieval instead of longer prompts, deterministic code instead of a call — which is exactly what Week 3 and Week 4 were teaching in miniature. And notice the 57× spread is not a rounding error to be optimised later: it is an **architecture** decision, because it determines which use cases are viable at all.

Q2 — You build the platform, but the pipeline is the bottleneck

Almost everyone starts with component 2 or 4, because a model call and a bit of infrastructure produce something demonstrable in a fortnight. The bottleneck is almost always component 1, and often component 3. Rent-ability is the clean way to see it. **Algorithm development: rentable today** — an API key and an afternoon. **Software infrastructure: rentable this quarter** — cloud, vector stores, orchestration, all commodity. **Experimentation platform: partly rentable**, but only if you have outcomes to measure, which most organisations do not instrument. **Data pipeline: not rentable at all**, because the thing that makes it valuable is precisely that it is yours — and it needs data owners to agree, systems that were never designed to talk to each other to talk to each other, definitions to be reconciled across departments that disagree, and a legal basis for the use. None of that is a technical problem, which is why technical teams consistently under-forecast it. The practical consequence, and the one worth arguing about in class: **the sequencing that produces demos and the sequencing that produces value are different sequences**. A pilot built on a hand-assembled extract will work and will not generalise, because the extract was the project. If you want one diagnostic question for a proposed AI initiative, it is not “which model?” — it is “*where does the data come from every single day, and who owns that path?*”

Q3 — It falls outside the operating-model framework, and that is the framework’s biggest weakness

Two claims have to be separated. The first is that a digital operating model dissolves the **internal** constraints on scale and scope — that one is not touched at all by what happened to Ant, and is if anything confirmed by it, since Ant reached a billion users with a fraction of a bank’s headcount precisely as described. The second is the implied strategic conclusion: that having dissolved those constraints, the firm keeps climbing. That is what the cancelled IPO falsifies, and it falsifies it in an instructive way. Ant was not stopped by a competitor, by a cost curve, or by a technology limit. It was stopped because a lending book of that size, intermediated by a platform that carried almost none of the credit risk, became a systemic-risk question — and systemic-risk questions get answered by regulators. So the honest reading is that **the framework accounts only for internal bottlenecks and is silent on external ones**, and the external ones are increasingly the binding constraint in exactly the industries where the framework works best: finance, healthcare, employment, insurance, anything touching personal data. There is a sharper version of this available to you in class, and it is worth arguing for: the appearance of a regulatory bottleneck is not random bad luck, it is **caused** by removing the internal ones. A firm that could only underwrite six loans a day per officer was never going to become systemically important. Scale is what made Ant a regulatory object. On that reading the framework does not just omit the bottleneck — it predicts it, and then declines to say so. Hold that thought for Capsule 3 — and hold the counter-cases with it, because one company in one country in one year proves very little.

Take two industries that removed the same internal bottlenecks and were never subject to a capital rule. **Social-media recommendation** built the purest version of the flywheel in the book: every click a free label, the ranking better for everyone each day, marginal cost of the next recommendation about nothing. No regulator stopped it — and an external constraint arrived anyway, as content, teen-safety and platform-transparency duties (the EU's Digital Services Act placed risk-assessment and transparency obligations on the largest platforms from 2023), and it arrived **because** ranking at that scale worked as well as it did. **Digital advertising** looked for two decades like the industry where no external constraint ever comes; then privacy rules and, more sharply, **Apple's App Tracking Transparency** — a change to its own operating system in April 2021 that required apps to ask permission before tracking users across other apps and websites — struck the data layer the flywheel actually ran on. **Meta said the change would cost it about US\$10 billion of 2022 revenue.** Note what that figure is: Meta's own estimate, made by the party with the strongest interest in the number being large, disputed at the time and never independently verified — interrogate it exactly the way Reading 2 teaches you to. **The mechanism is the point, not the number: the constraint was imposed by a supplier Meta could not do without, and no regulator was involved.** So the generalisation is broader and more uncomfortable than “regulated industries get regulated”. **The missing term is not regulation. It is any externally imposed constraint that appears because you succeeded** — and it can be imposed by a regulator, by a platform you depend on, or by the public. Come to class ready to say which of those three your own use case is most exposed to, because the framework will not tell you.

Q4 — Most use cases are buying scale, and pretending to buy learning

The three tests are deliberately ordered so that they get harder to pass. Scale is the easiest to establish and to defend, but it is worth being precise about what a pass looks like. The weak pass is **linear**: ten times the volume costs ten times as much today, automating the unit decision creates real value, and that is a perfectly respectable answer. The strong pass is **superlinear** — cost rises faster than volume because supervision, error rates, queueing and hiring lead time compound on top of headcount, and the process degrades before it becomes unaffordable. Rooms systematically under-claim this one, because the extra cost sits in other people's budgets and in quality, not in the line item they are looking at. The fail is a curve that is **already flat**, which is not a verdict on the use case, only a signal that the argument has to be made on scope or learning instead. Scope is a design question rather than a technology one — it asks whether the asset you are about to build has a **second internal customer**, and it is where the biggest returns quietly sit, because the second use case rides on the first one's pipeline. Learning is where the claims outrun the evidence. The test is not “does it collect data” — everything collects data. It is **does using the system produce a correct label, for free, that flows back in and changes behaviour.** Credit and fraud pass outright: default and chargeback are labels the world supplies without being asked, which is why financial services was the first industry to industrialise this and why the case study you will discuss is a lending business. Recommendations pass, because a click is a label. Most customer-service, drafting and summarisation use cases **fail**: the outcome is an opinion, nobody records it, and the only way to obtain a label is to pay a human to produce one — which is a cost line, not a flywheel. Applying this honestly to your own example is uncomfortable and it is the point. A use case that buys only scale is still a good use case; it just does not compound, and it should be costed and defended as an efficiency project rather than sold internally as a strategic moat. Getting that distinction right in front of a board is most of what Capsule 1 is training you to do.

Q5 — Advantage lives in what takes years, not in what takes an afternoon

Run the cull. By next Tuesday a competitor can have: the same frontier model, the same agent framework, the same cloud, the same vector database, a prompt very close to yours, and — if you published a demo — your general approach. That is the entire technology stack, gone in a week. What survives the cull tends to be four things. **Proprietary data with a lawful basis to use it**: not merely data you hold, but data you have consent, provenance and permission for — this is the one that is measured in years and it is why the pipeline is component one. **Distribution**: being where the decision already happens. Alipay's advantage in lending was never a superior model, it was that the loan offer appeared at the checkout the customer was already at. **Workflow integration and switching cost**: once your system is the place work is done, replacing it means retraining people and rewiring processes, which is a cost measured in disruption rather than dollars. And **regulatory permission**: licences, approvals and accreditations that took three years to obtain and cannot be bought. Rank them and the ordering is roughly regulatory permission > proprietary data > distribution > workflow integration, though the last three are close and industry-dependent. Two implications worth carrying into your project. First, if your idea's only advantage is the model, you do not have an advantage — you have a feature, and it has the shelf life of the next release. Second, and more usefully: this list is a **checklist for use-case selection**, not just a post-hoc explanation. Prefer the use case that sits on data you already lawfully hold, inside a workflow you already own, in a place where you already have permission to operate. That is a boring answer, and it is the one that survives contact with a budget.

APPENDIX — STRATEGY VOCABULARY, EXPLAINED

The distinction that carries this reading: the business model is what you promise and how you get paid; the operating model is how the work actually gets done — and it is the operating model that sets the ceiling. Every term below is a way of being precise about the second one.

Business model

- **Plain.** What you sell, to whom, and how the money reaches you.

Technically. The value proposition plus the revenue and cost architecture that captures part of that value. Two firms with identical business models can have completely different economics — which is the article's opening move.

Operating model

- **Plain.** How the work gets done, and what has to happen for you to do twice as much of it.

Technically. The configuration of people, processes, technology and data that delivers the value proposition. Iansiti and Lakhani's central claim is that AI changes the operating model, and that the operating model — not the business model — is what limits scale, scope and learning.

AI factory

- **Plain.** A repeatable production line that turns data into decisions, the way a factory turns steel into cars.

Technically. Four components: **data pipeline**, **algorithm development**, **experimentation platform**, **software infrastructure**. It is a factory because the output is manufactured at a known cost per unit on a line that runs continuously — not because the technology is industrial.

Data pipeline

- **Plain.** The plumbing that gets the right data to the model, correctly and on time, every time — without anyone doing it by hand.

Technically. Systematic, sustainable, scalable gathering, cleaning, integration and processing of data. The only one of the four components that cannot be rented, and the reason most AI pilots do not become AI systems.

Experimentation platform

- **Plain.** The machinery for finding out whether a change actually caused the improvement.

Technically. Controlled testing — A/B tests, holdouts, staged rollouts — at a cadence the organisation can afford. The same logic as the Week-3 eval set, one level up: without a comparison you cannot separate improvement from drift, seasonality or luck.

Scale, scope, learning

- **Plain.** More of the same thing • the same capability used somewhere else • getting better because you ran it.

Technically. The three sources of value in a digital operating model. Their tests: at 10× volume does the cost curve stay linear or bend upward, and does 10× volume exist (scale); does the asset have a second internal customer (scope); does use generate a free, correct label that flows back (learning). Learning is the one most often claimed and least often true.

Network effects — direct, indirect and data

- **Plain.** The product is better because other people use it — or because other people's use trains it.

Technically. **Direct:** value rises with the number of same-side users (a payment network). **Indirect / cross-side:** value rises with the other side of a marketplace (merchants and consumers). **Data network effects:** usage improves the *predictions* for everyone. Only the third depends on AI, and it requires the free-label test to be passed.

Collision

- **Plain.** A firm built as software walks into your industry and the fight is not fair — not because it is better, but because it is built differently.

Technically. The encounter between a digital and a traditional operating model. The entrant typically wins the segments the incumbent's cost per transaction made unservable, then expands upward. It cannot be answered by adopting a tool, only by changing the operating model.

Fixed versus variable cost, in an AI system

- **Plain.** What you pay once, versus what you pay again every single time somebody uses it.

Technically. Fixed: pipeline, platform, infrastructure, and any model you trained yourself. Variable: per-token inference, retrieval, and human review. In 2020 the second bucket was small; in 2026 it is the cost of goods sold, and it is why AI-native gross margins run **50–60%** against **80–90%** for classic software. Reading 3 makes this the whole subject.

SOURCE LINKS

Companion readings in this pack — these are the “Reading 2 / Reading 3 / Reading 4” references used throughout this brief. **Reading 2** — McKinsey, *The State of AI* (adoption against value, and the six per cent who see it at the EBIT line). **Reading 3** — *AI unit economics and cost to serve* (the arithmetic behind every cost claim made here). **Reading 4** — *GraphCast, Climate TRACE and AI in verticals* (the two archetypes of a use case). **Case pre-read 1** — *Ant Group (A)*, the Capsule-3 case, and this article's own flagship example.

Iansiti & Lakhani — “Competing in the Age of AI”, Harvard Business Review, Jan–Feb 2020 • hbr.org/2020/01/competing-in-the-age-of-ai (optional — paywalled; NTU students can read it via the Library’s Business Source Complete)

Book page — Competing in the Age of AI (HBR Press, 2020), Harvard Business School •

www.hbs.edu/faculty/Pages/item.aspx?num=56633 (optional — buy or borrow; Ch. 1–2 are the same argument at length)

Bessemer Venture Partners — “The AI pricing and monetization playbook” (10 Feb 2026) • www.bvp.com/atlas/the-ai-pricing-and-monetization-playbook (the 50–60% vs 80–90% gross-margin figures)

Companion • **Reading 2** supplies the frequency this article’s exemplars cannot; **Reading 3** prices the marginal-cost claim; the **case pre-read** is this article’s own flagship example, five years later.

CAVEATS

1. **This brief is the assigned reading and is written to stand alone**, because the article is paywalled and the University’s licence does not let us circulate it. It is still a summary: the article carries the authors’ own voice and their fuller examples, and if you can reach it through the Library it is worth the fifteen minutes. Note also that the article is itself the *compressed* version of the book — where the book gives cases and detail, the article gives the skeleton, and the skeleton is what we use.
2. Published January 2020. Its examples pre-date foundation models entirely, and its flagship example was stopped by regulators ten months later. The mechanism has aged well; the calibration has not.
3. The evidence base is selected exemplars — firms chosen because they succeeded — so it supports claims about mechanism and not about how often the mechanism pays off. Reading 2 is the corrective.
4. The Ant figures quoted here are those printed in the 2020 article and in our Capsule-3 case. Ant’s valuation, ownership and regulatory position have all changed substantially since; the case pre-read carries the update.
5. This summary was generated and fact-checked using Anthropic models. The fixed/variable decomposition, the “free label” test, the rent-ability ranking and the four criticisms in “Where the argument has aged” are the instructor’s reading, not the authors’. Flag anything that looks wrong.